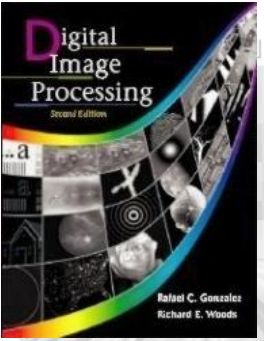

Chapter 6
Color Image Processing

The use of color in image processing is motivated by two principal factors. First, color is a powerful descriptor that often simplifies object identification and extraction from a scene. Second, humans can discern thousands of color shades and intensities, compared to about only two dozen shades of gray. This second factor is particularly important in manual (i.e., when performed by humans) image analysis.

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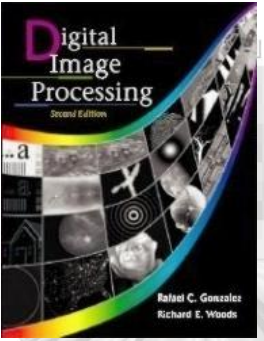

Color Fundamentals

- When color is available, it gives much more information about an image than intensity alone.
- Color is very useful for recognition of objects in an image both for humans and computers.

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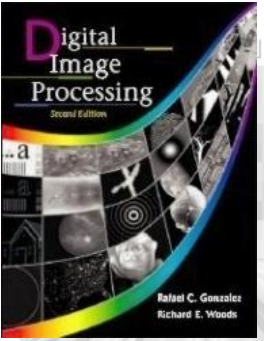

Color Fundamentals

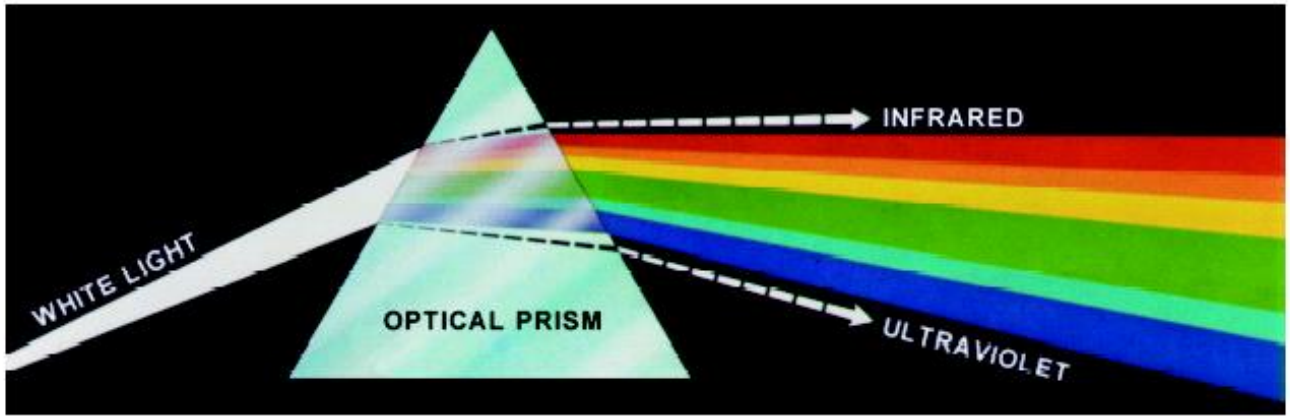

FIGURE 6.1 Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lamp Business Division.)

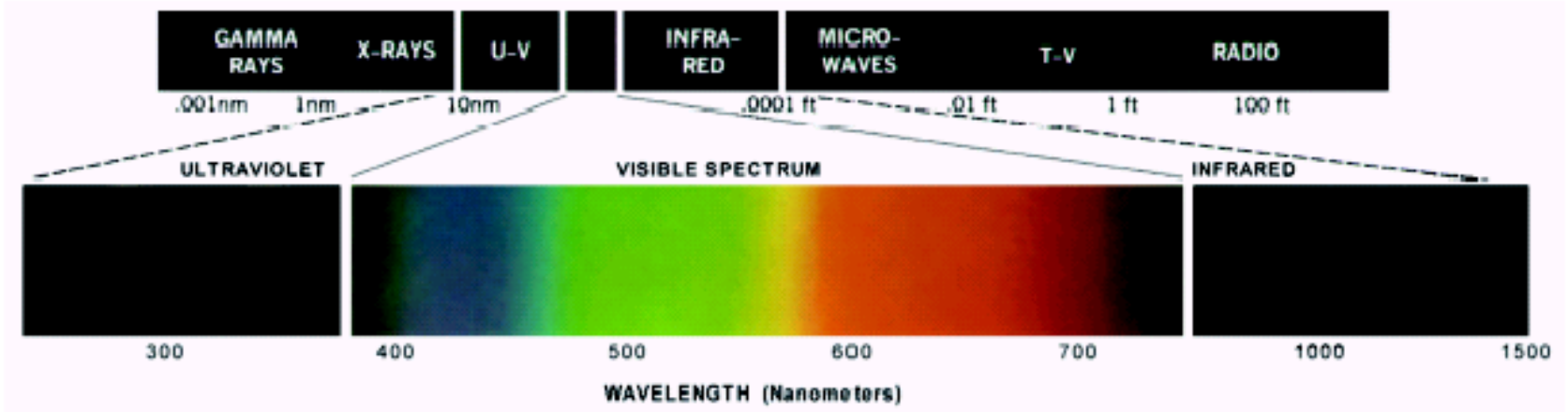

FIGURE 6.2 Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lamp Business Division.)

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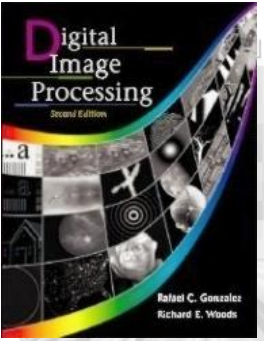

Color Fundamentals

- The actual color perceived by a human of an object depends on both the color of the illumination and the reflectivity of the object, as well as the sensitivity of human perception.
- Objects appear to be different colors because they absorb and reflect different colors of light. A blue object, for example, reflects blue light while absorbing other colors.
- Grey objects or grey images reflect and absorb all frequencies of light about equally, so they do not appear colored.

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Color Fundamentals

- Color is sensed by the eye using three kinds of cones, each sensitive primarily to red, green or blue, though there is significant overlap.
- We refer to red, green and blue as the primary colors, and denote to set as RGB.

blue = 435.8nm

green = 546.1nm

red = 700nm

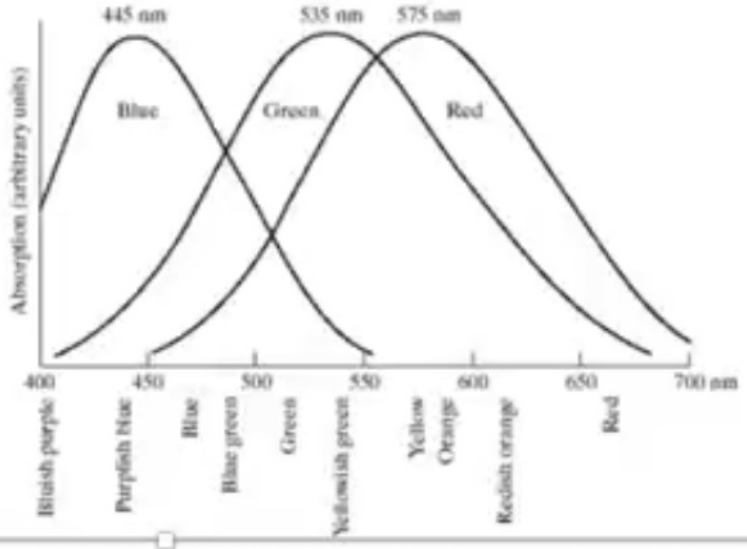

FIGURE 6.3 Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

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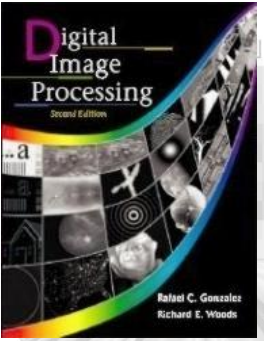
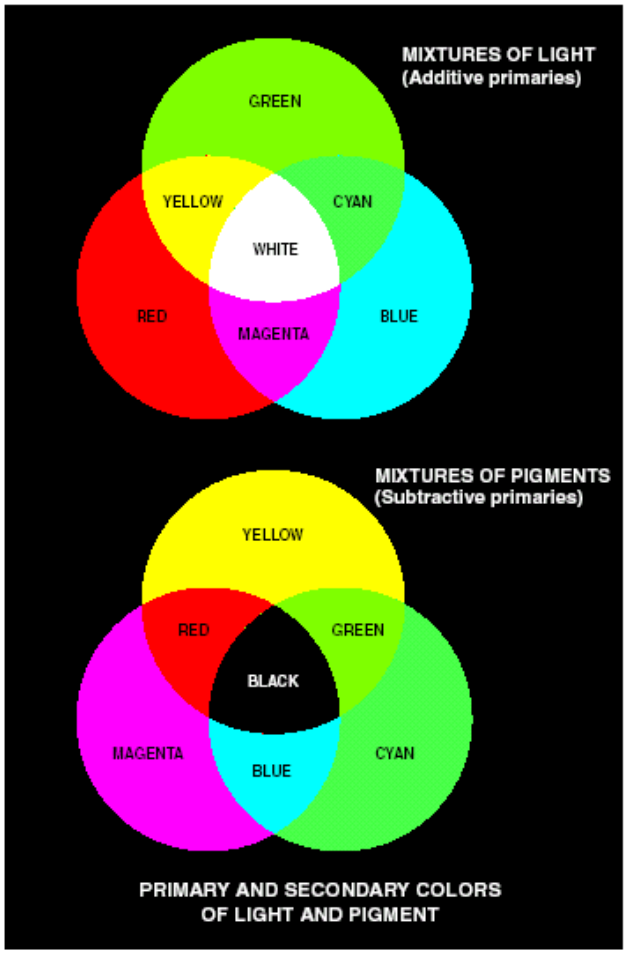


Color Fundamentals

- Secondary colors: magenta (red + blue), cyan (green + blue), and yellow (red + green)



a
b

FIGURE 6.4 Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lamp Business Division.)

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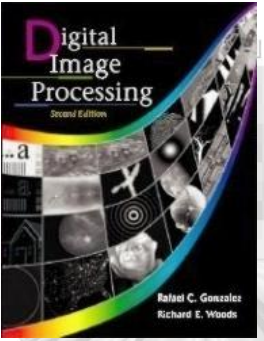

Chromaticity Diagram

- The amounts of red, green, and blue needed to form any particular color are called tristimulus values (X, Y and Z). These represent three dimensional coordinates of any perceived color.
- The tristimulus values can be normalized to give trichromatic coefficients, x (red), y (green) and z (blue). Note that because of normalization, $x+y+z=1$.
- Since x, y and z are not independent, only x and y are enough to specify a color.
- If the wavelength of the pure colors are plotted in these coordinates, and the mixtures of these wavelengths are plotted inside the pure colors, the result is known as the CIE chromaticity diagram.
- In the chromaticity diagram, white light is defined as the mixture of equal amounts of all wavelengths of visible light.

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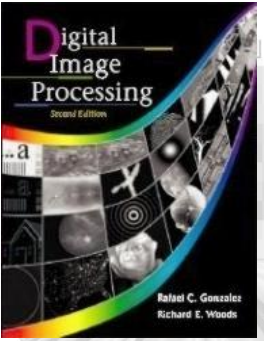
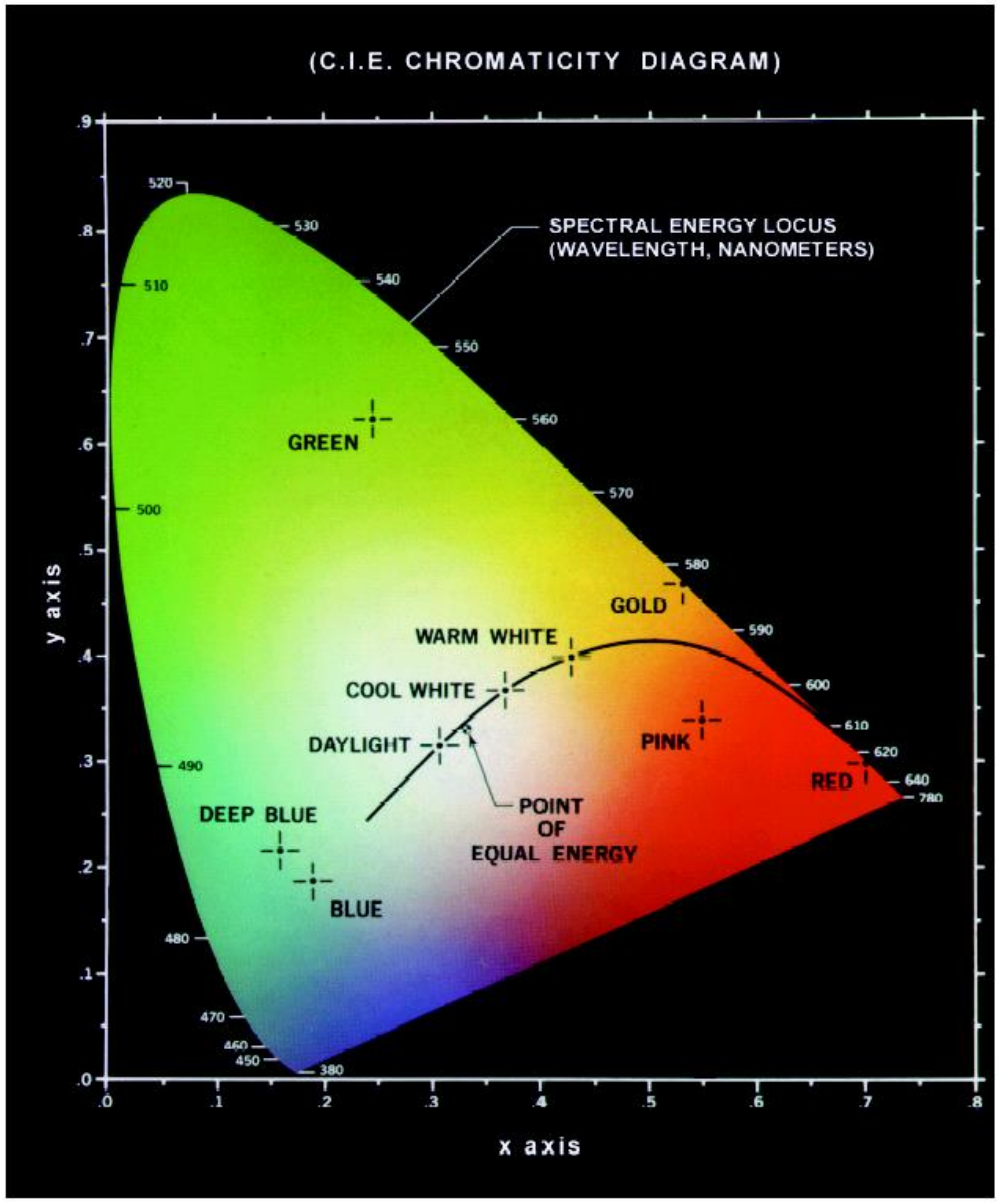


Chromaticity Diagram

FIGURE 6.5 Chromaticity diagram. (Courtesy of the General Electric Co., Lamp Business Division.)



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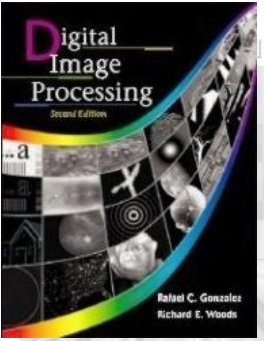

Color Fundamentals

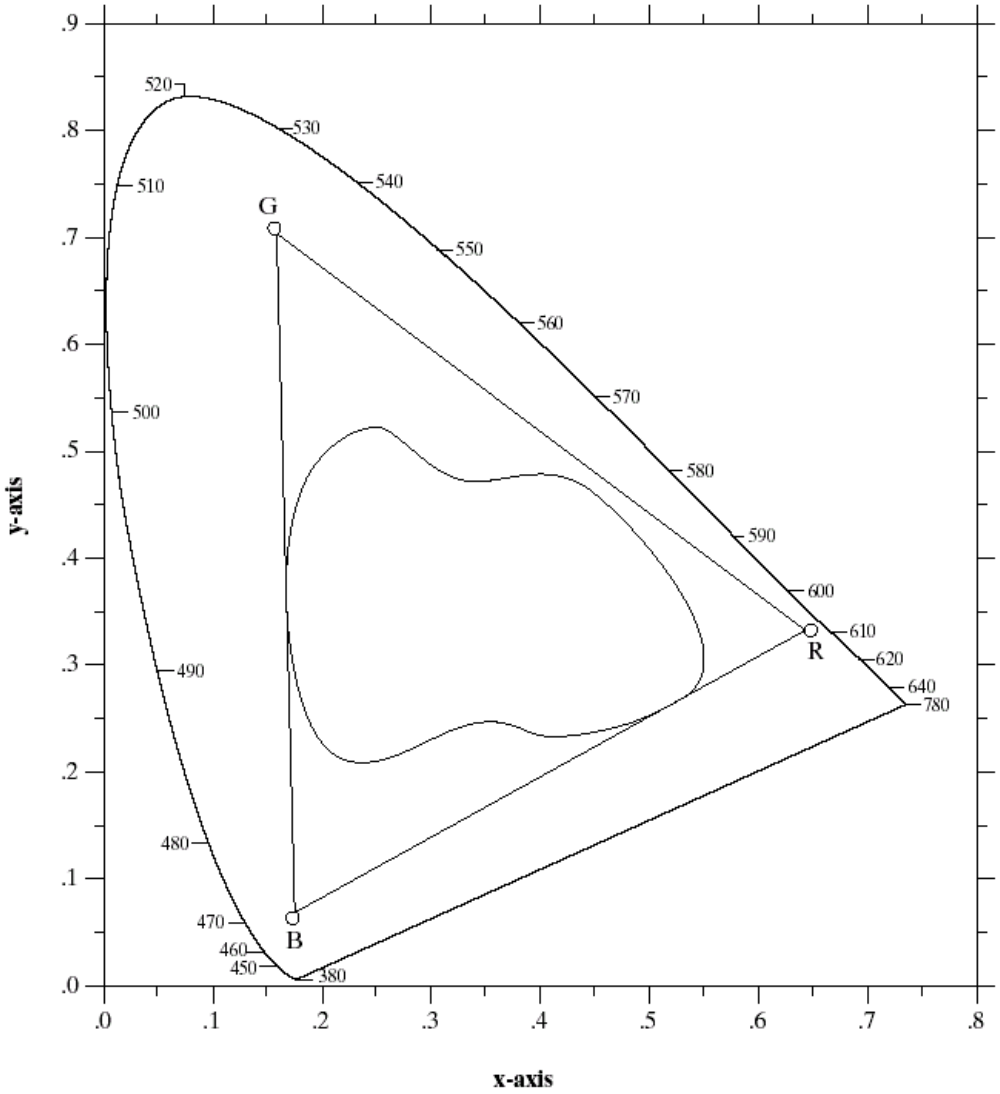

FIGURE 6.6 Typical color gamut of color monitors (triangle) and color printing devices (irregular region).

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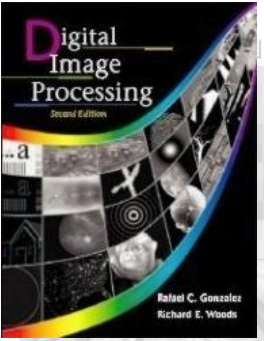
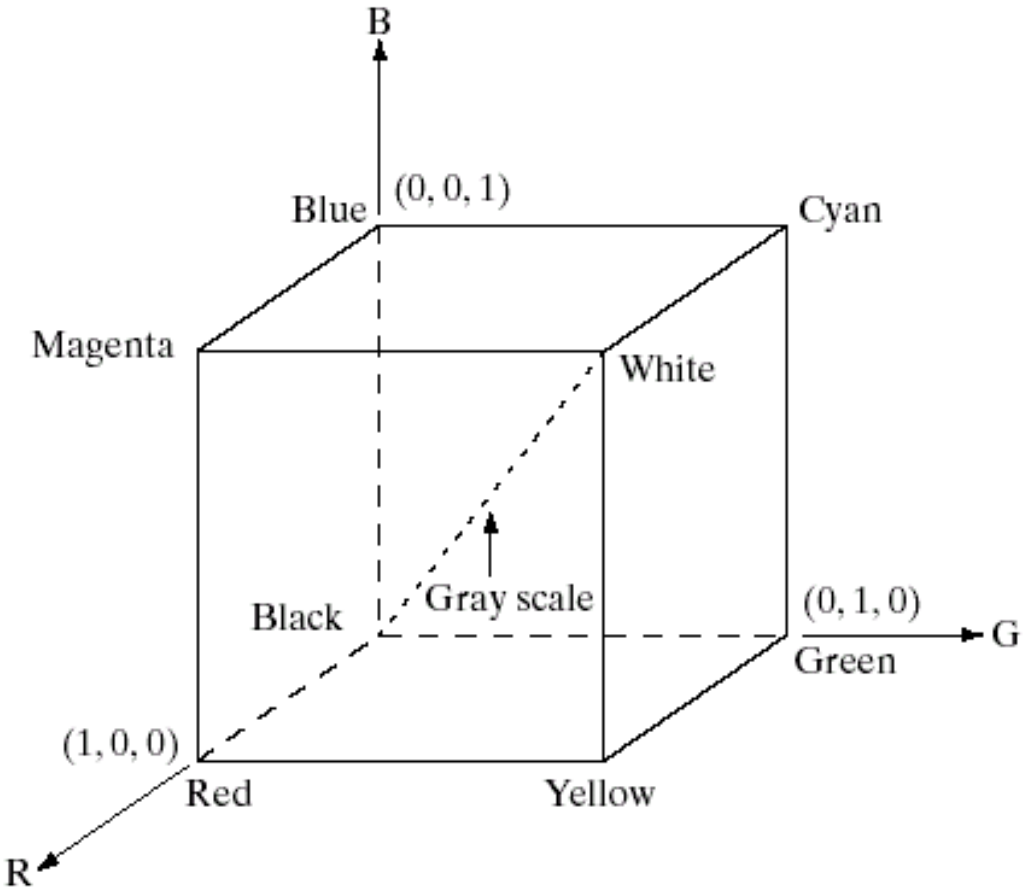


Color Models -- RGB Model

FIGURE 6.7
Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point (1, 1, 1).



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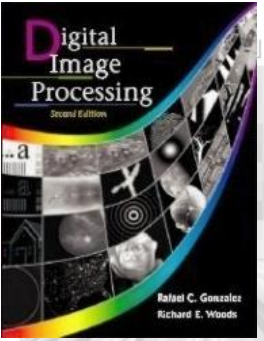

Color Models -- RGB Model

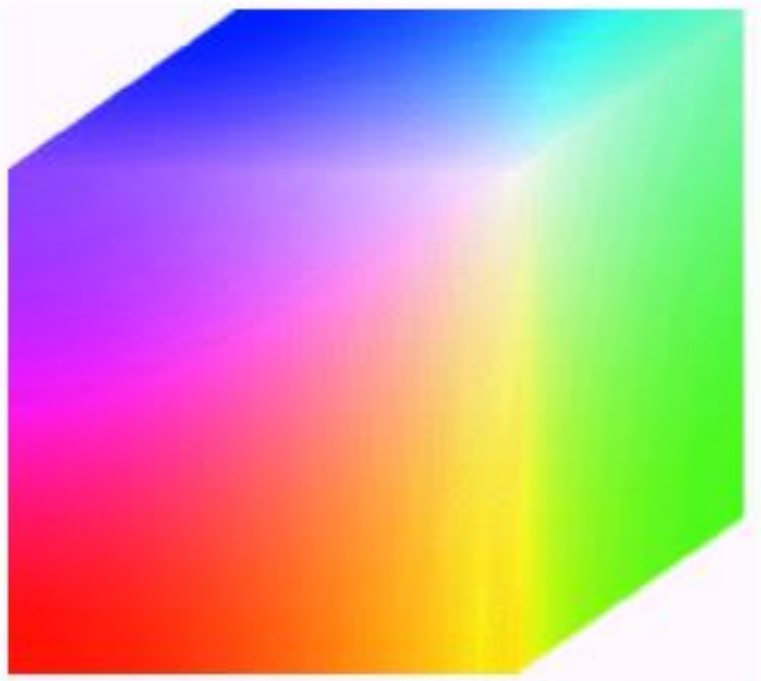

FIGURE 6.8 RGB 24-bit color cube.

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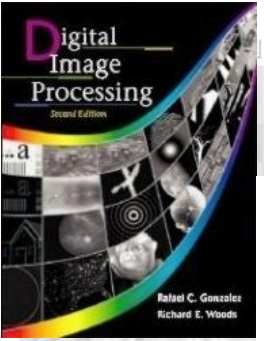
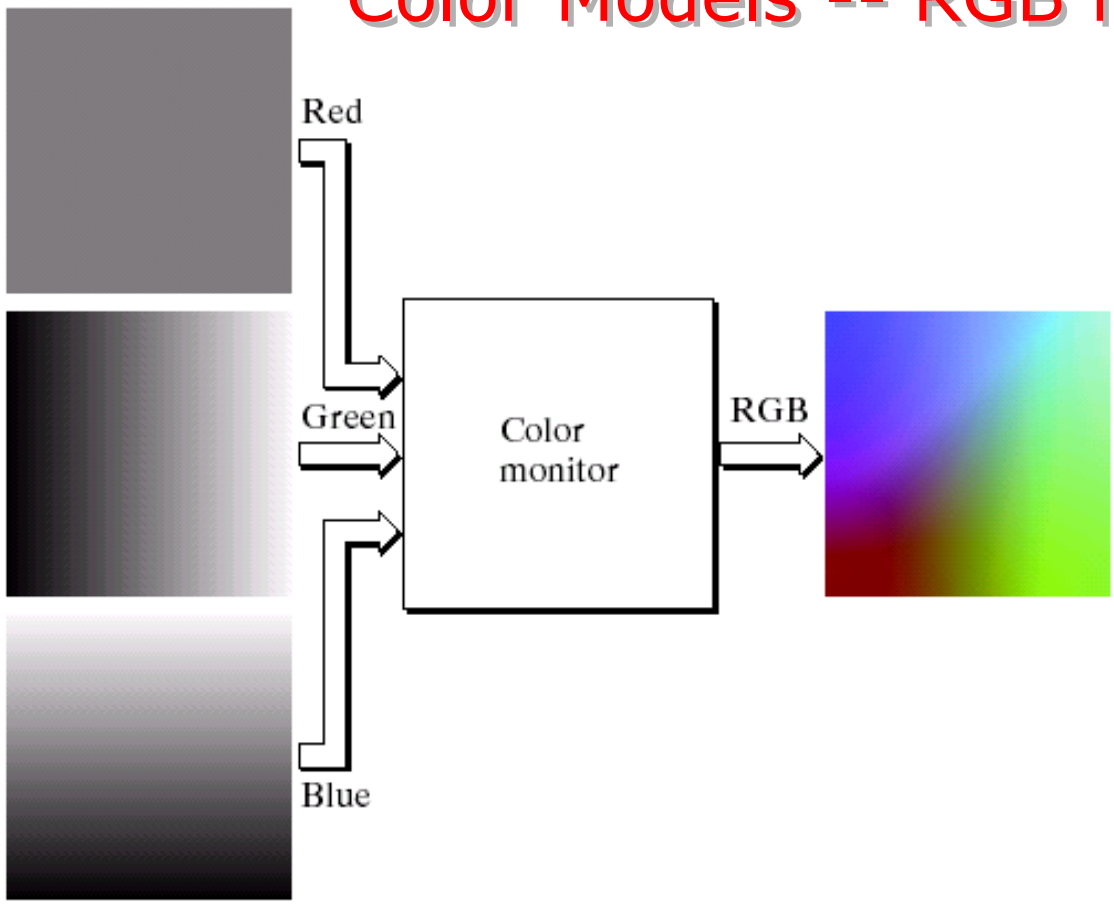
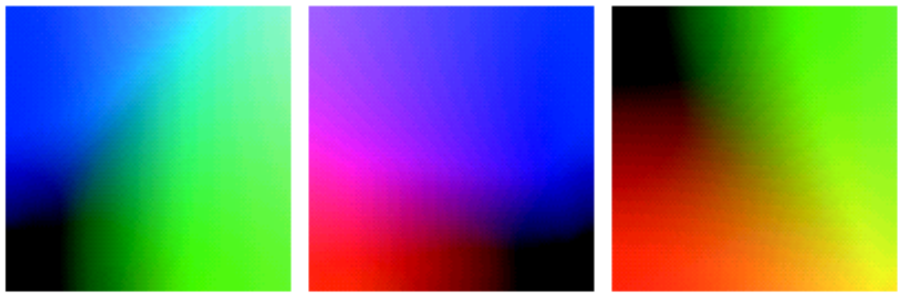


Color Models -- RGB Model

a
b

FIGURE 6.9
(a) Generating the RGB image of the cross-sectional color plane (127, G, B).
(b) The three hidden surface planes in the color cube of Fig. 6.8.





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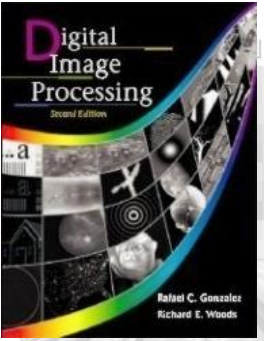
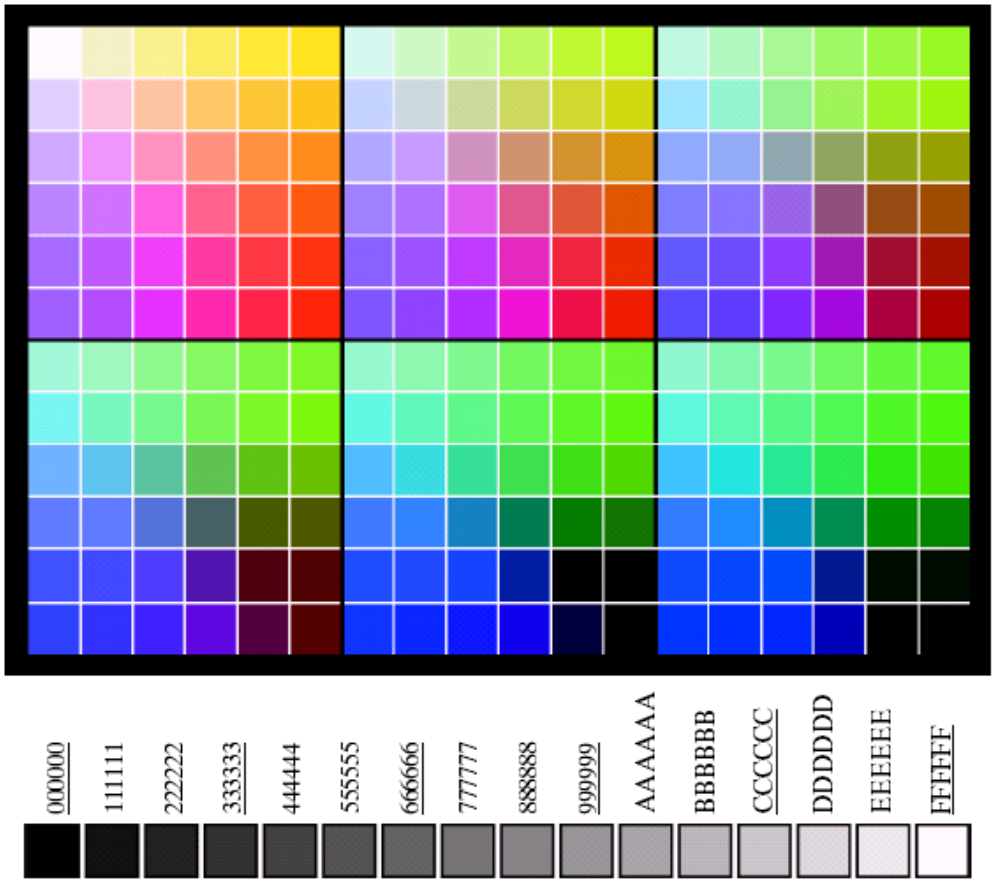


Color Models -- RGB Model

- For most graphics images used for Internet applications, a set of 216 colors has been selected to represent "safe colors" which should be reliably displayed on computer monitors.

Number System	Color Equivalents					
Hex	00	33	66	99	CC	FF
Decimal	0	51	102	153	204	255

TABLE 6.1
Valid values of each RGB component in a safe color.



a
b

FIGURE 6.10
(a) The 216 safe RGB colors. (b) All the grays in the 256-color RGB system (grays that are part of the safe color group are shown underlined).

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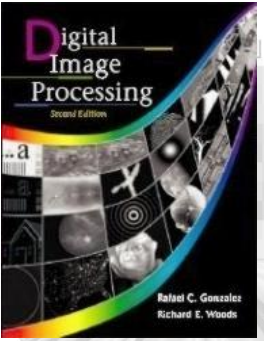

Color Models -- RGB Model

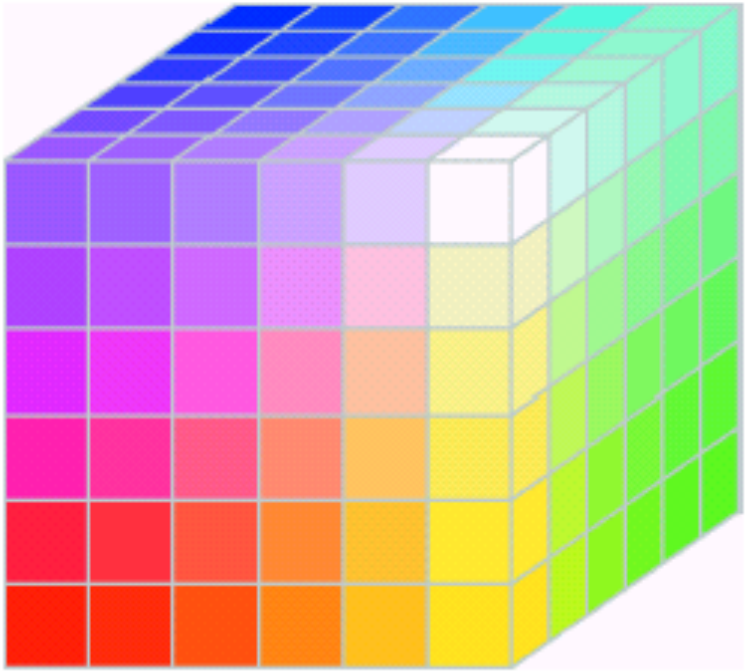

FIGURE 6.11 The RGB safe-color cube.

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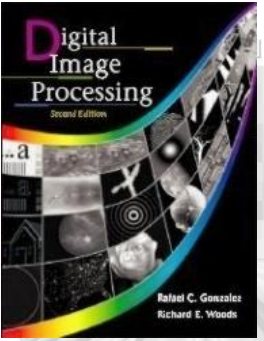
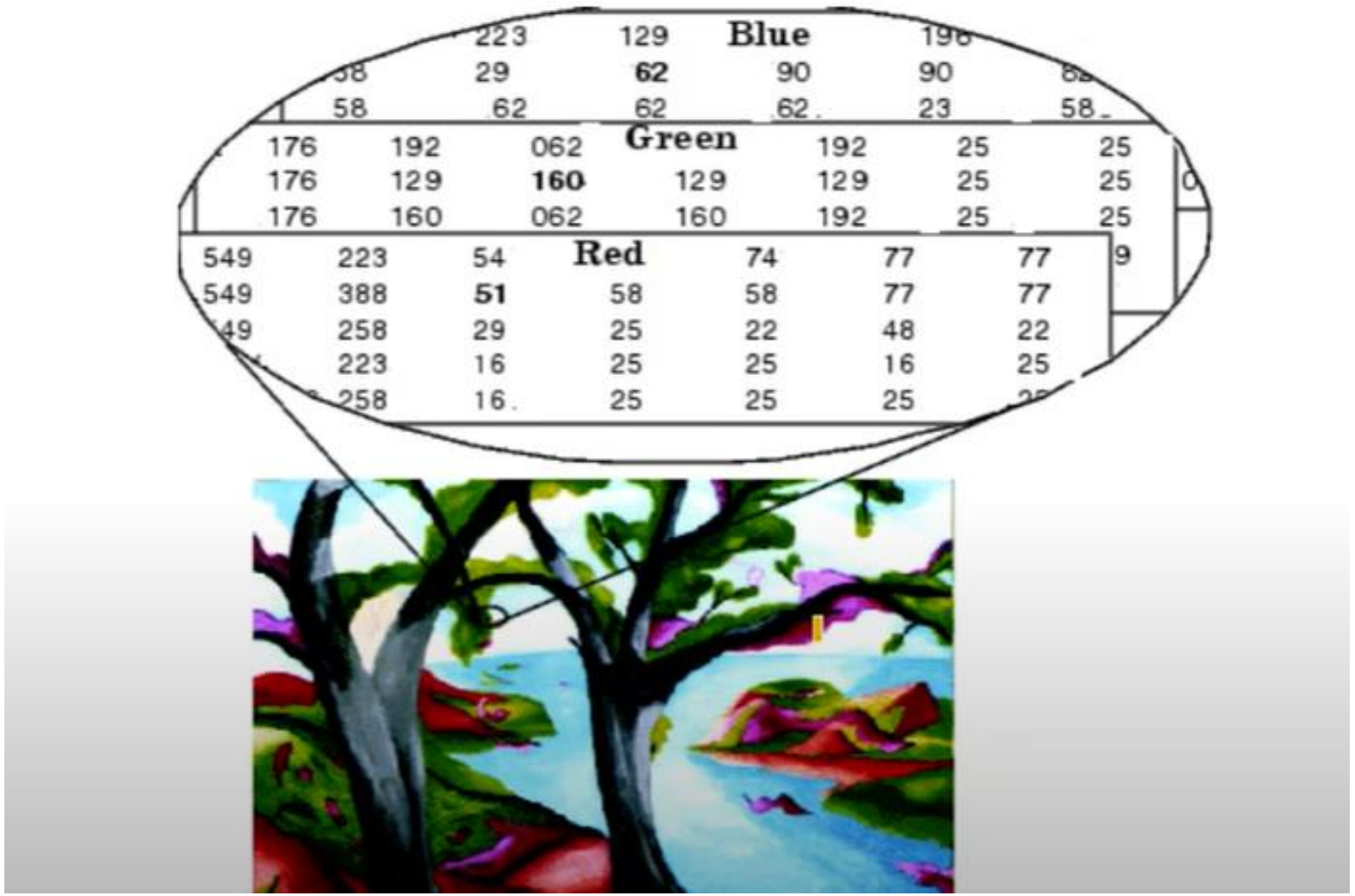


Color Models -- RGB Model



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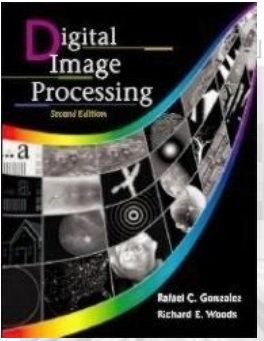

Color Models -- CMY and CMYK Models

- An RGB to CMY conversion

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

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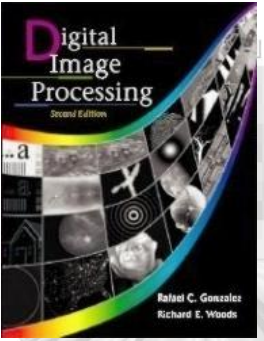
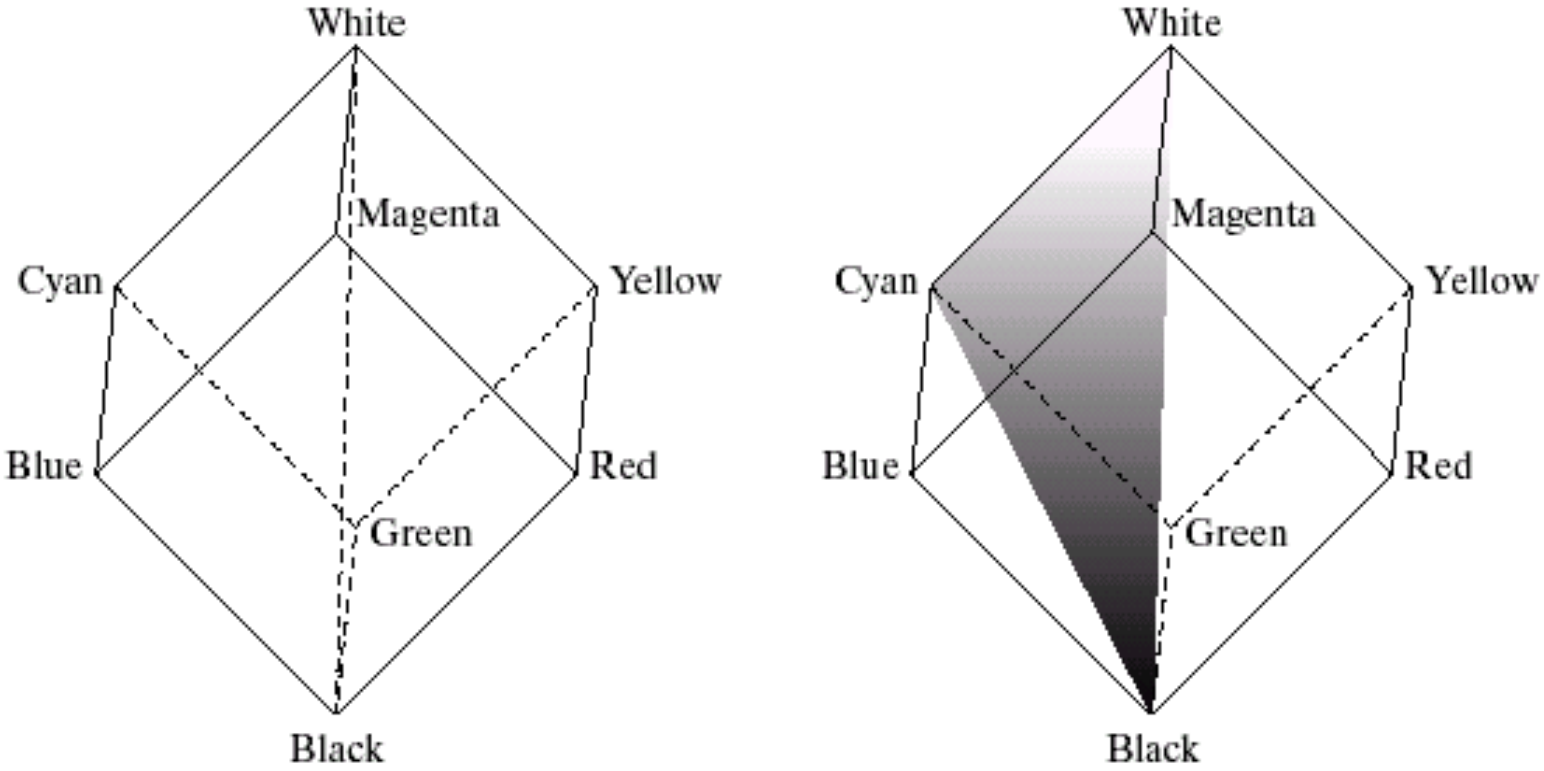


Color Models -- HSI Model



a b

FIGURE 6.12 Conceptual relationships between the RGB and HSI color models.

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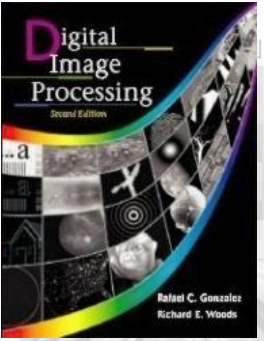
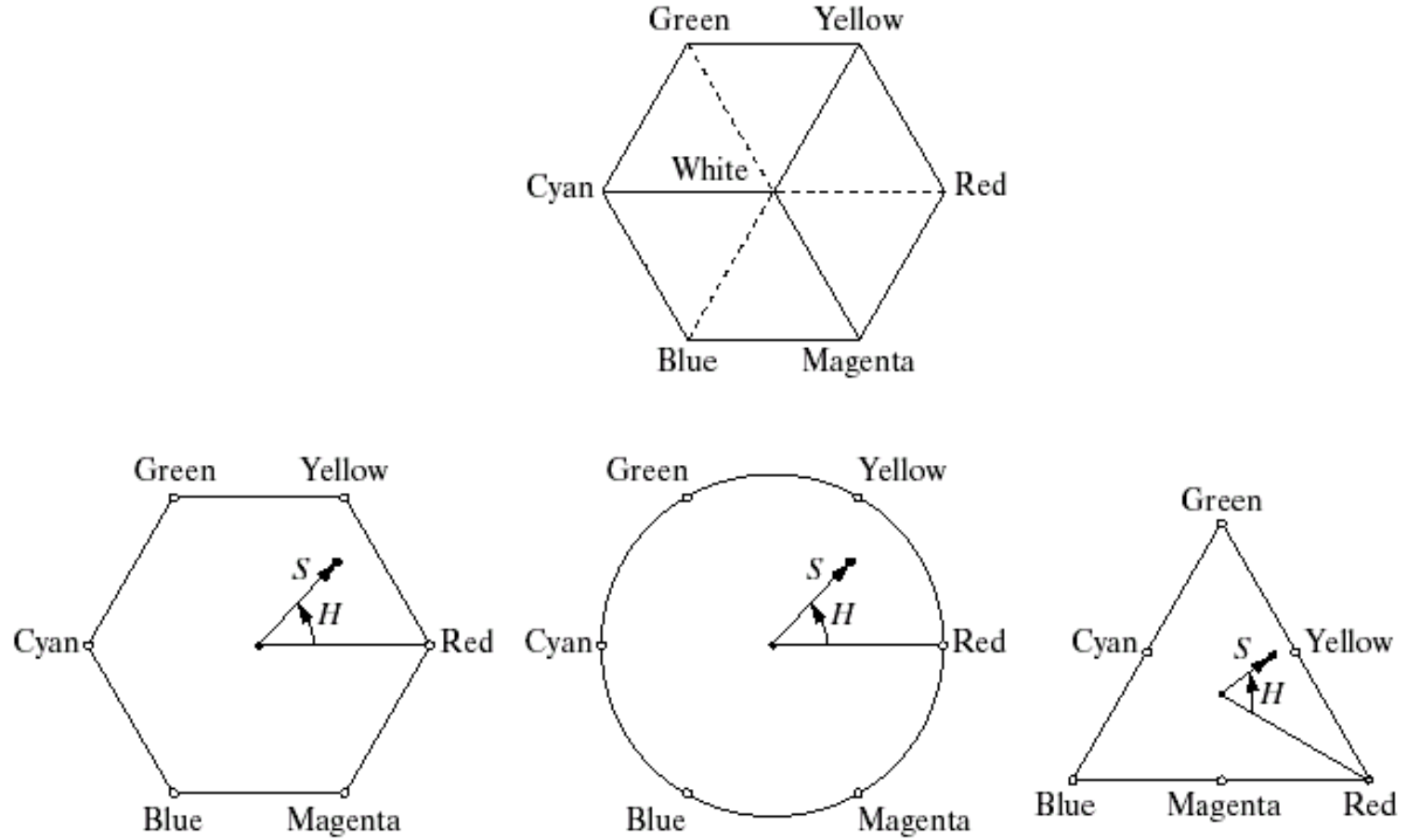


Color Models -- HSI Model



a
b c d

FIGURE 6.13 Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation. The intensity of all colors in any of these planes is given by the position of the plane on the vertical intensity axis.

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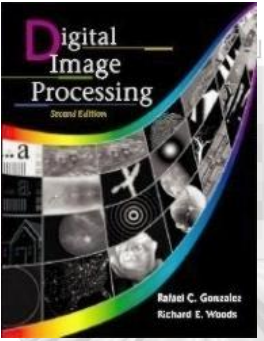
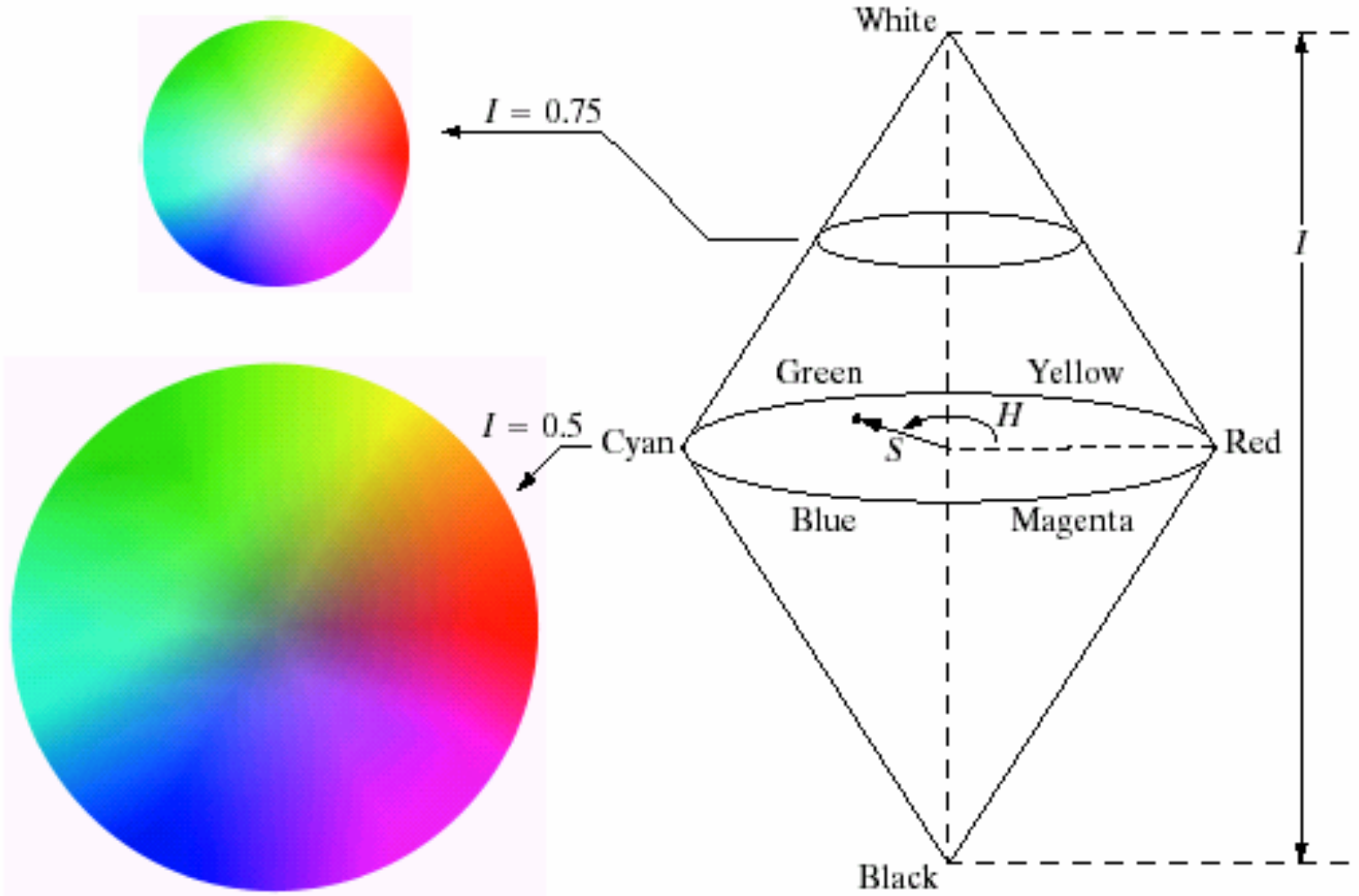


Color Models -- HSI Model



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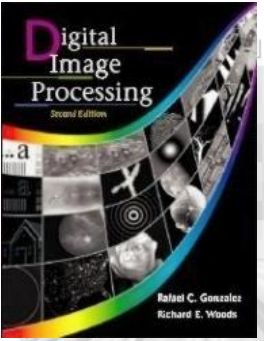
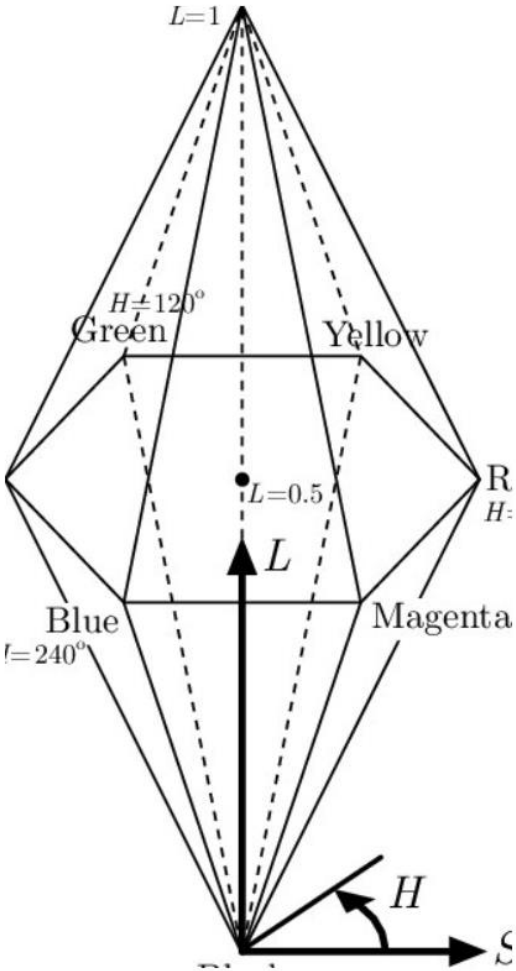


Color Models -- HSI Model



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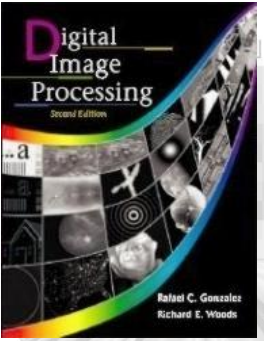

Color Models -- HSI Model

- Converting colors from RGB to HSI

$$H = \begin{cases} \theta & \text{if } B \le G \\ 360 - \theta & \text{if } B > G \end{cases} \qquad \theta = \cos^{-1}\left\{ \frac{\frac{1}{2}\left[(R-G)+(R-B)\right]}{\left[(R-G)^2+(R-B)(G-B)\right]^{\frac{1}{2}}} \right\}$$

$$S = 1 - \frac{3}{(R+G+B)}\left[\min(R,G,B)\right]$$

$$I = \frac{1}{3}(R+G+B)$$

H ⟵ زاوية بعد اللون عن محور

S ⟵ تشبع اللون

I ⟵ شدة اللون

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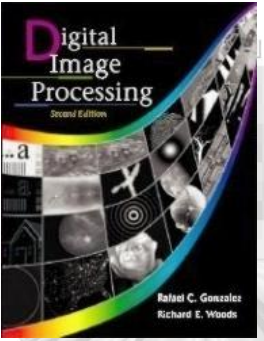

Color Models -- HSI Model

- Converting colors from HSI to RGB
  - *RG* sector ($0^o \le H < 120^o$)

$$B = I(1-S)$$

$$R = I\left[1+\frac{S\cos H}{\cos(60^o - H)}\right]$$

$$G = 3I-(R+B)$$

  - *GB* sector ($120^o \le H < 240^o$): see page 300.
  - *BR* sector ($240^o \le H \le 360^o$): see page 300.

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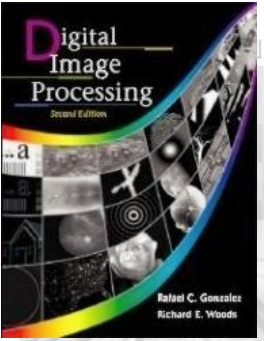
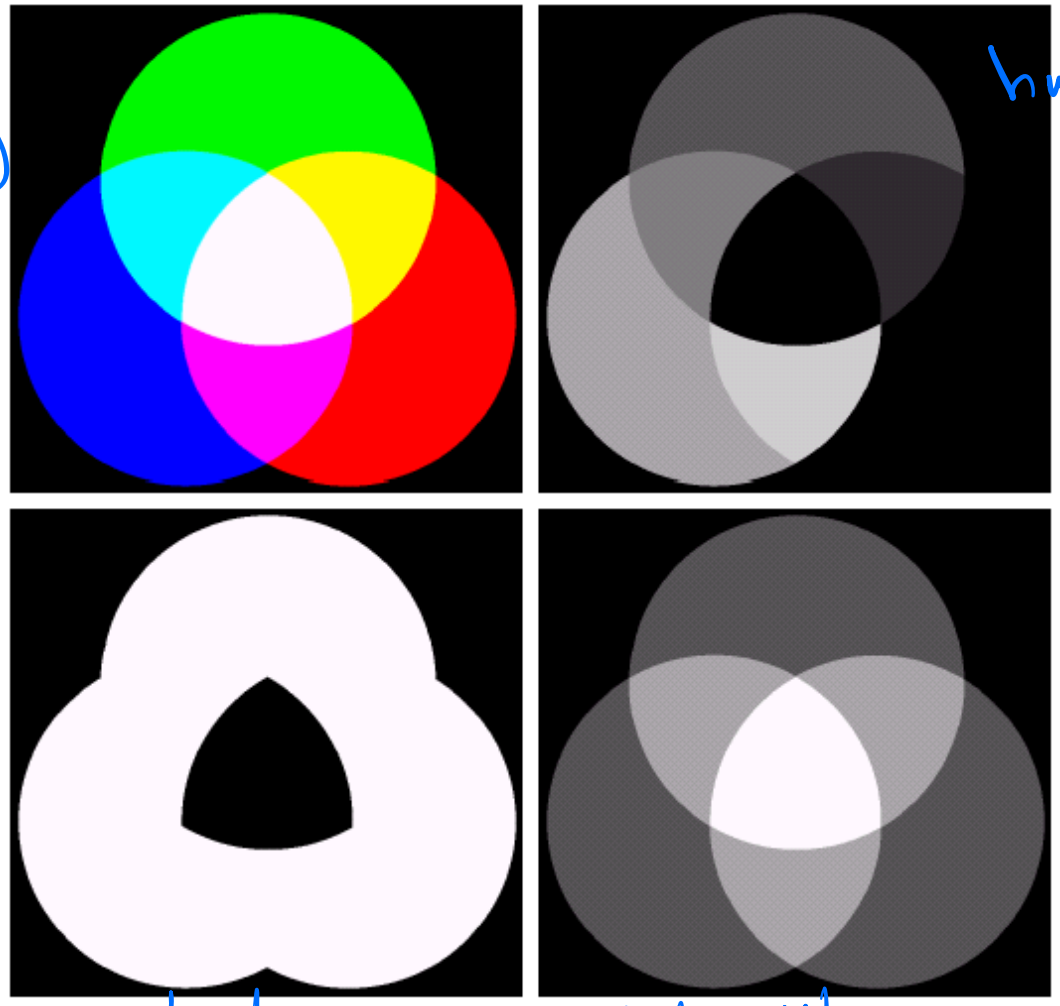


Color Models -- HSI Model

```
rgb = imread('papers.png');
hsv = rgb2hsv(rgb);
imshow(hsv(:,:,1));
imshow(hsv(:,:,2));
imshow(hsv(:,:,3));
```



a b
c d

FIGURE 6.16 (a) RGB image and the components of its corresponding HSI image: (b) hue, (c) saturation, and (d) intensity.

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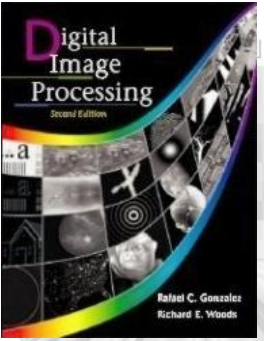

Pseudocolor Image Processing
Intensity Slicing

- The pseudocolors used to colorize a gray level image do not represent the original, true colors if there were originally a color image.

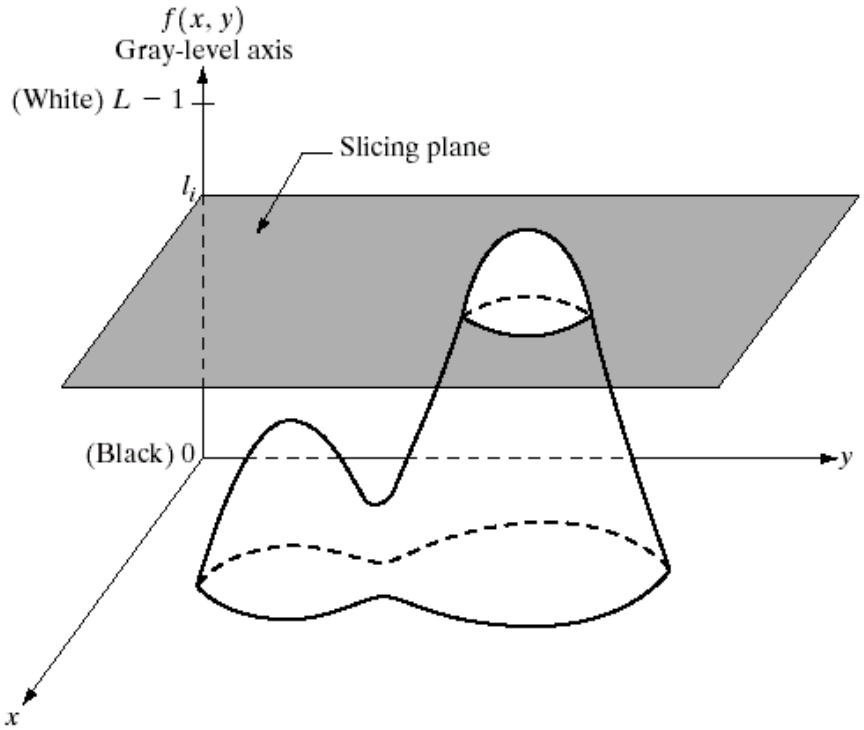

FIGURE 6.18 Geometric interpretation of the intensity-slicing technique.

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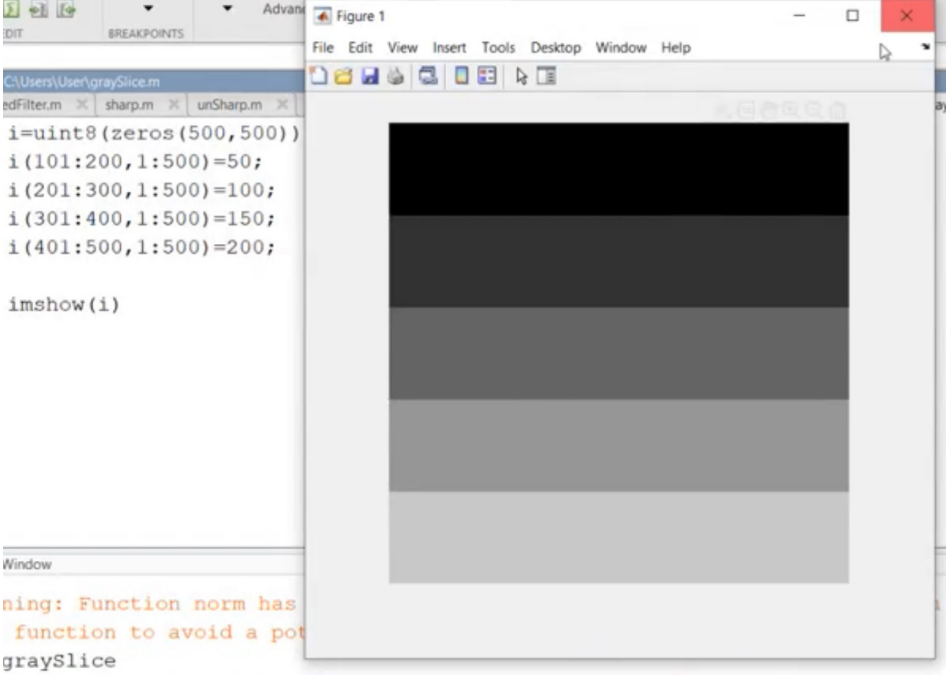
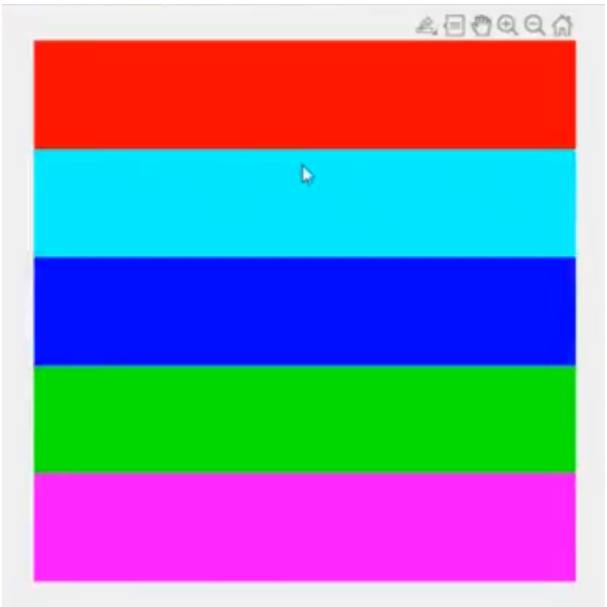
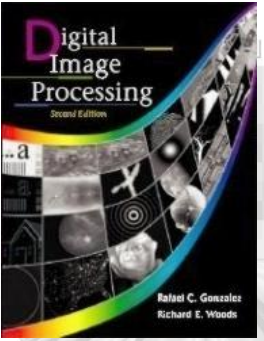


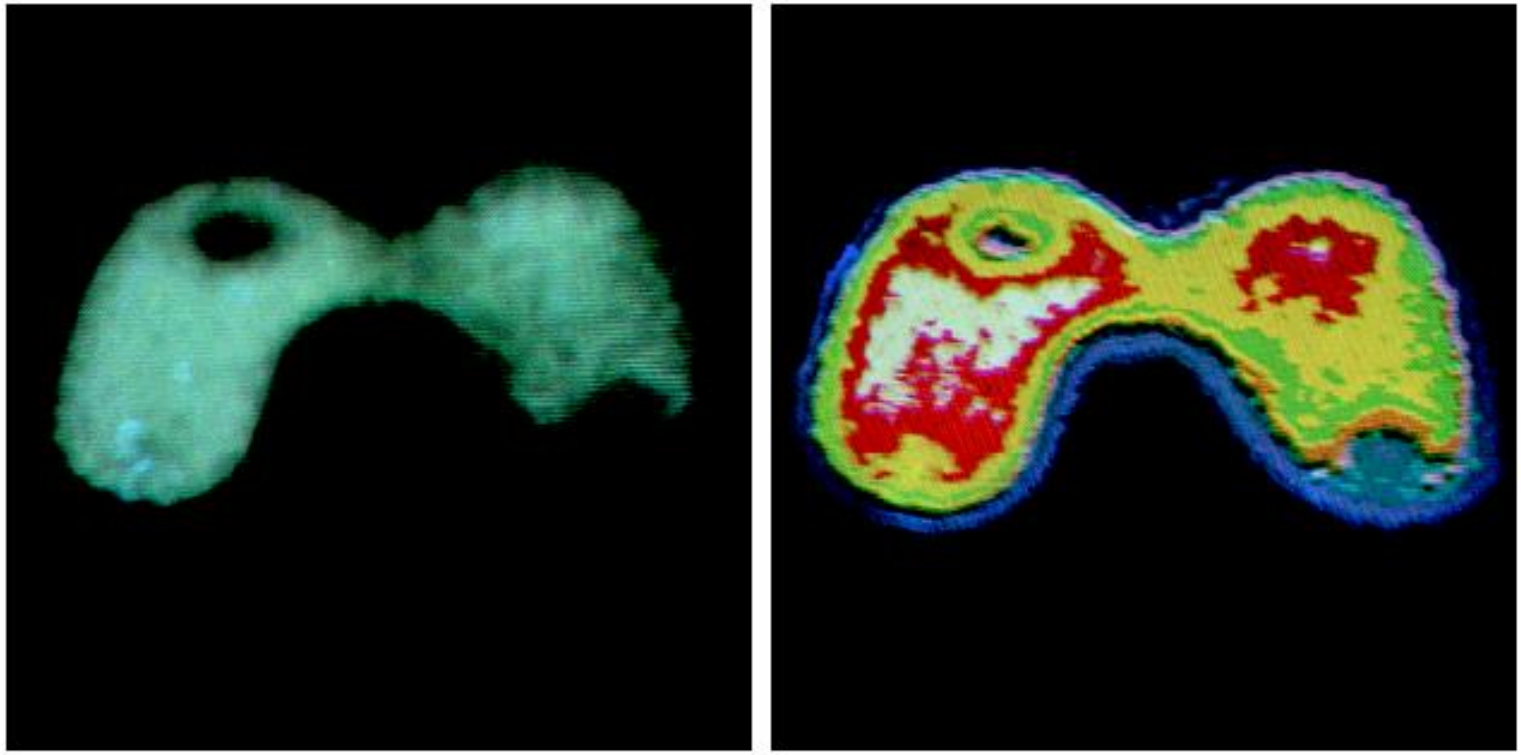

```
i=uint8(zeros(500,500));
i(101:200,1:500)=50;
i(201:300,1:500)=100;
i(301:400,1:500)=150;
i(401:500,1:500)=200;
r=[1 0 0;0 1 1;0 0 1;0 1 0;1 0 1];
j=grayslice(i,6);


imshow(j,r);
```



```
i=imread('snowflakes.png');
imshow(i);
r=[1 0 0;0 1 1;0 0 1;0 1 0;1 0 1;1 1 0;1 1 1];
j=grayslice(i,16);
imshow(j,r)
```



Pseudocolor Image Processing
Intensity Slicing



a b

FIGURE 6.20 (a) Monochrome image of the Picker Thyroid Phantom. (b) Result of density slicing into eight colors. (Courtesy of Dr. J. L. Blankenship, Instrumentation and Controls Division, Oak Ridge National Laboratory.)

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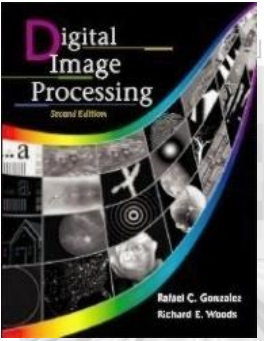

Full Color Image Processing

- Two processing methods:
  - (1) process each channel (or color component) separately, as if the color image were three gray scale images;
  - (2) process all channels with each pixel represented as a vector.

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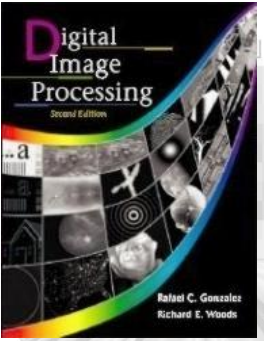
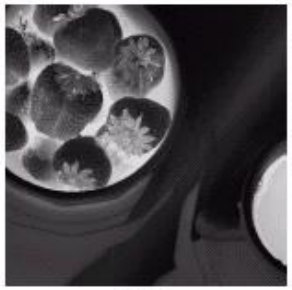


Color Transformations RGB<->HSI<->CMYK

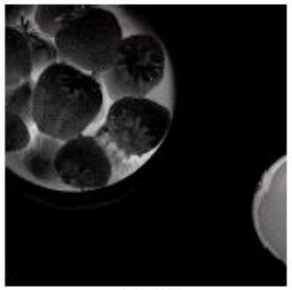
Full color

FIGURE 6.30 A full-color image and its various color-space components. (Original image courtesy of Med-Data Interactive.)


Cyan

Magenta

Yellow


Black

Red

Green

Blue

Hue

Saturation

Intensity

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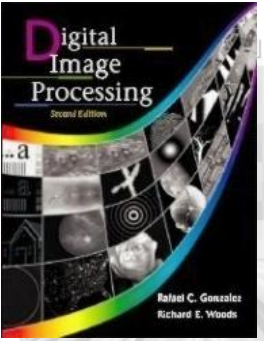
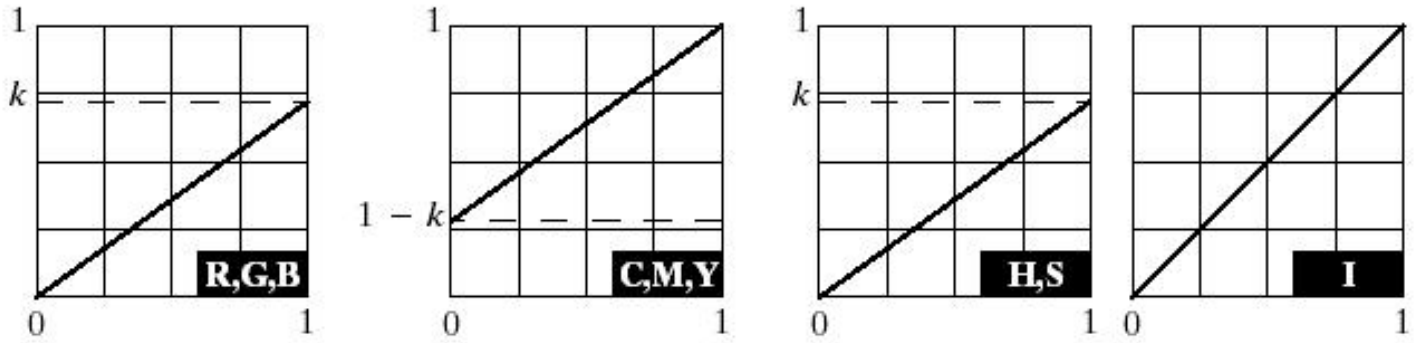


Color Transformations Example

a b
c d e

FIGURE 6.31 Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c)–(e) The required RGB, CMY, and HSI transformation functions. (Original image courtesy of MedData Interactive.)



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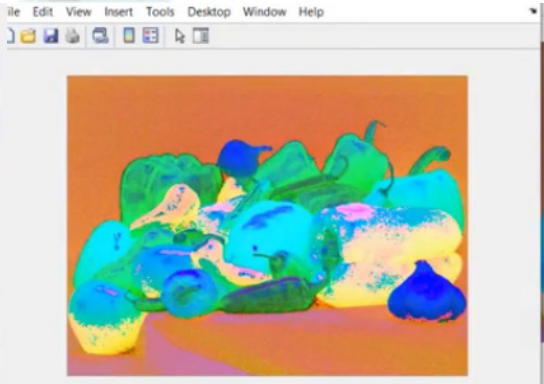
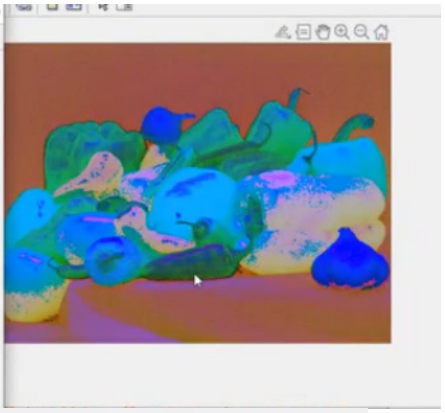
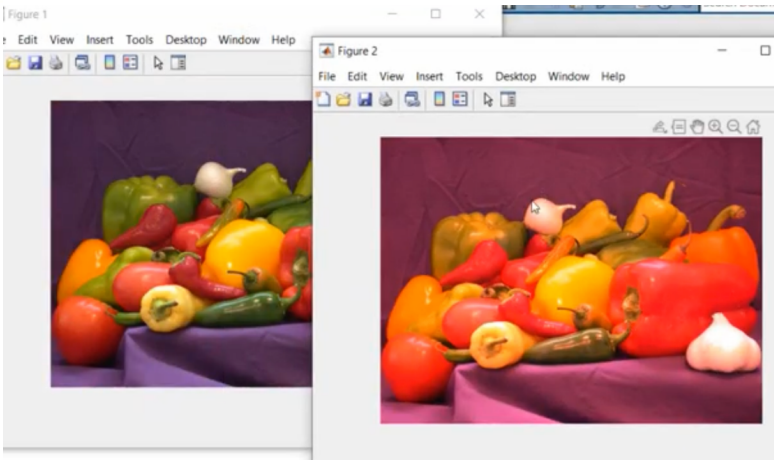
```
i=imread('peppers.png');
j=rgb2hsv(i);
imshow(i);figure;imshow(j);
h=j(:,:,1).*0.7;
s=j(:,:,2).*0.7;
v=j(:,:,3);
j_pro=cat(3,h,s,v);
figure;imshow(j_pro);
```



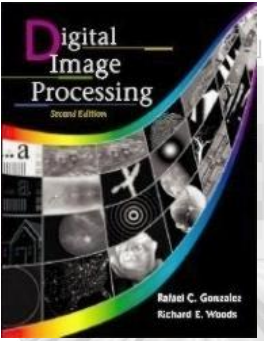


```
i=imread('peppers.png');
imshow(i);
r=i(:,:,1).*0.7;
g=i(:,:,2).*0.7;
b=i(:,:,3);
j_pro=cat(3,r,g,b);
figure;imshow(j_pro);
```

```
i=imread('peppers.png');
imshow(i);
r=i(:,:,1).*1.7;
g=i(:,:,2);
b=i(:,:,3);
j_pro=cat(3,r,g,b);
figure;imshow(j_pro);
```




```
i=imread('peppers.png');
imshow(i);
r=i(:,:,1).*0.04;
g=i(:,:,2);
b=i(:,:,3);
j_pro=cat(3,r,g,b);
figure;imshow(j_pro);
```



Color Complement Transformations

- To the human visual system, colors have complements.
- Complements are basically given by subtracting one color from white, or by changing a hue by 180 degrees.

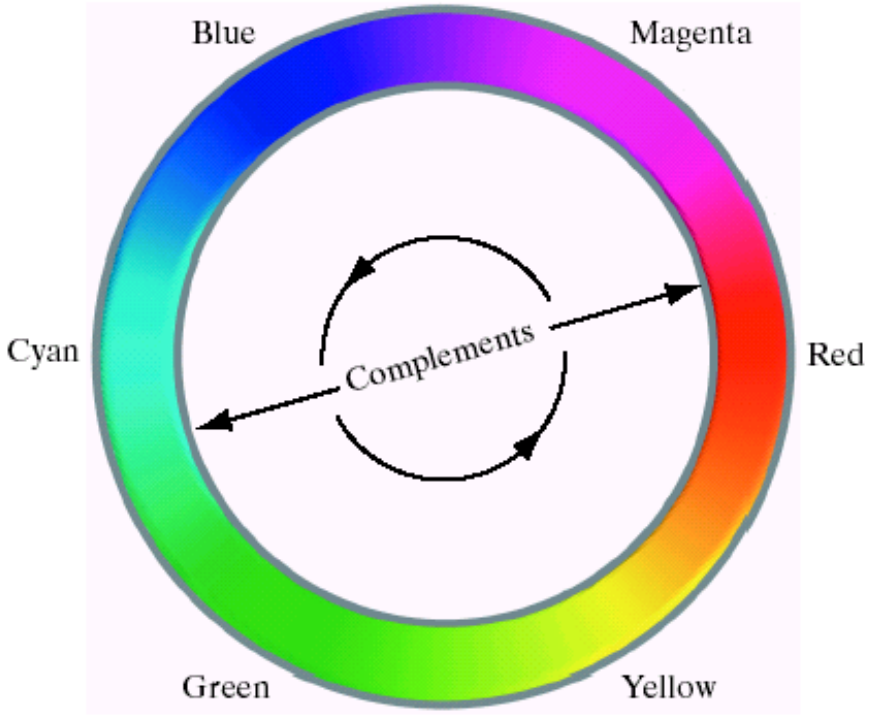


FIGURE 6.32
Complements on the color circle.

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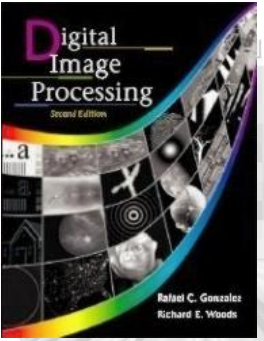
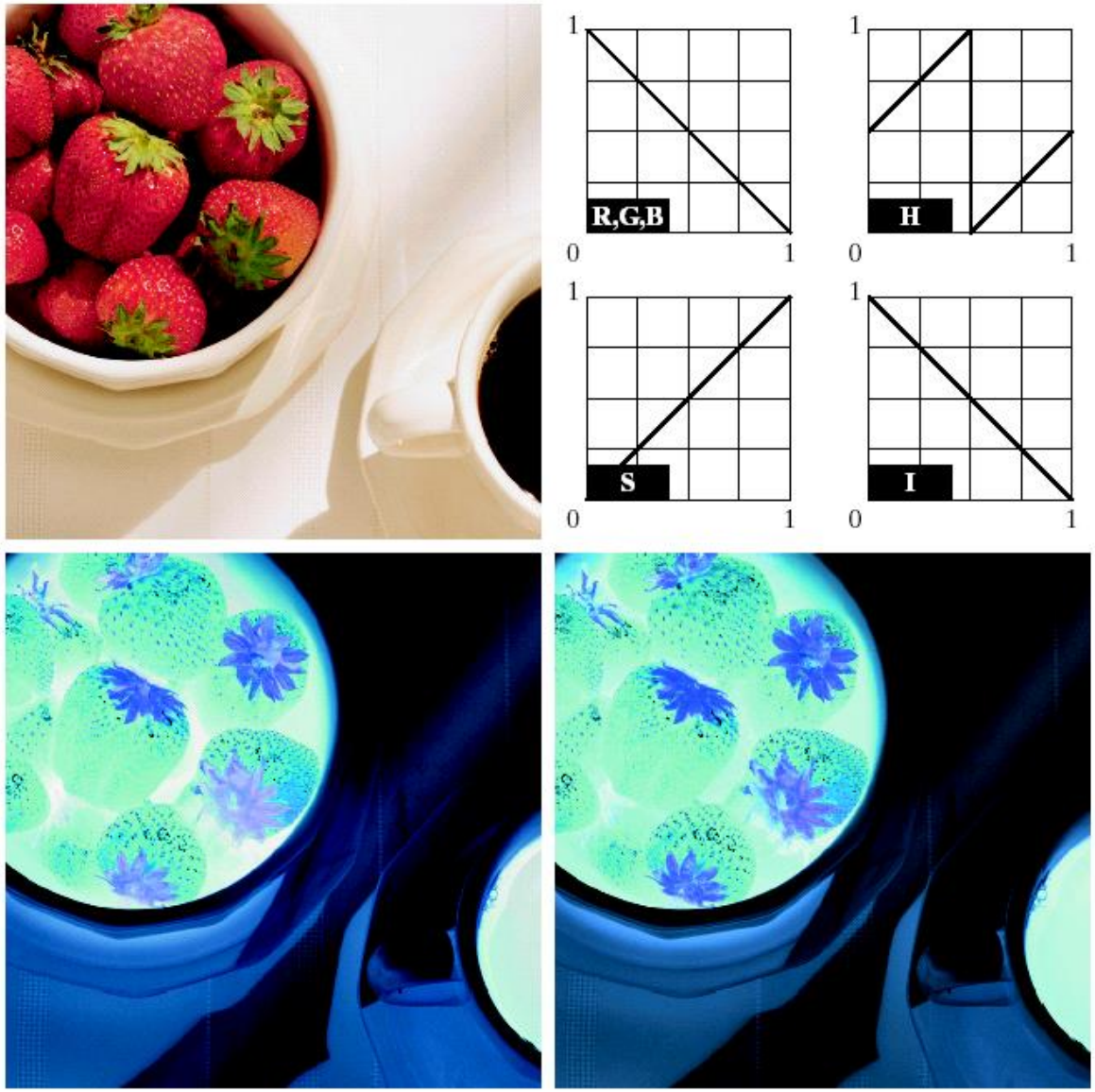


Color Complement Transformations Example



a b
c d

FIGURE 6.33 Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.

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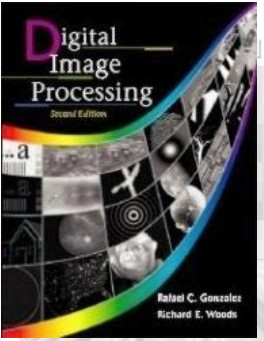
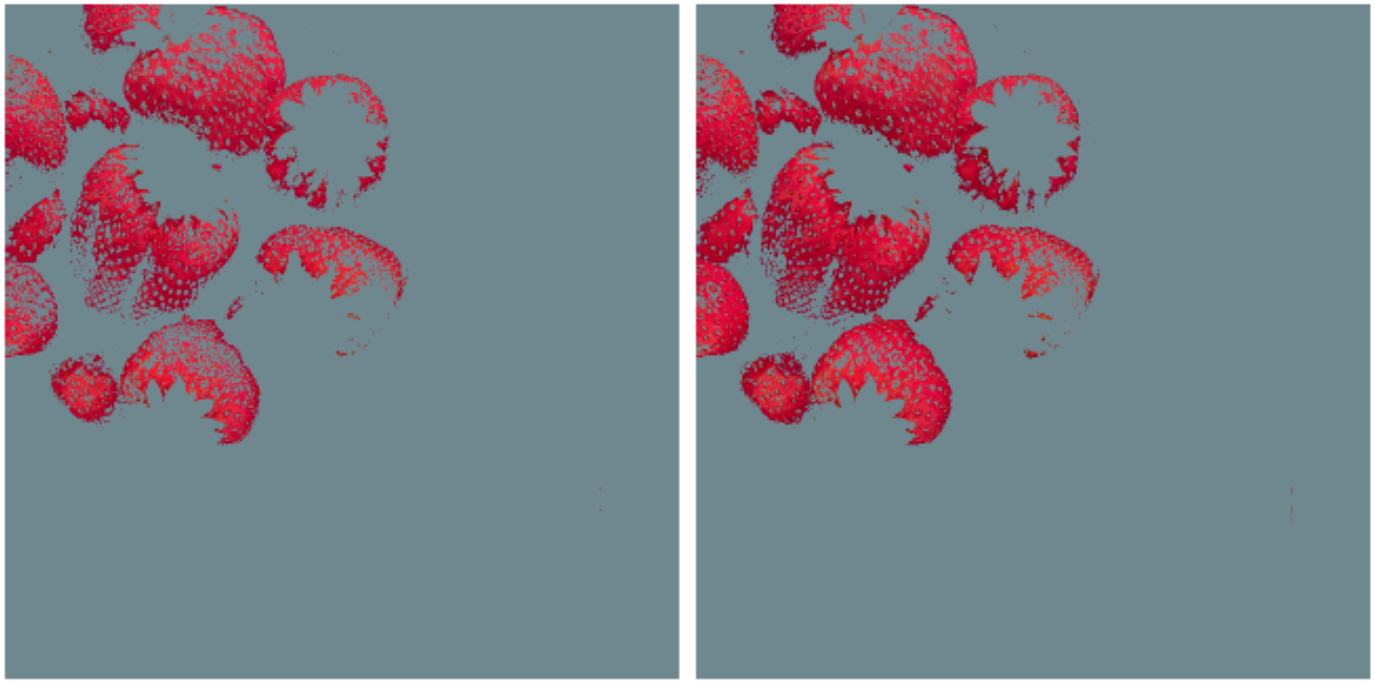


Color Slicing Example

- Color slicing: highlighting a specific range of colors in an image.



a b

FIGURE 6.34 Color slicing transformations that detect (a) reds within an RGB cube of width $W = 0.2549$ centered at (0.6863, 0.1608, 0.1922), and (b) reds within an RGB sphere of radius 0.1765 centered at the same point. Pixels outside the cube and sphere were replaced by color (0.5, 0.5, 0.5).

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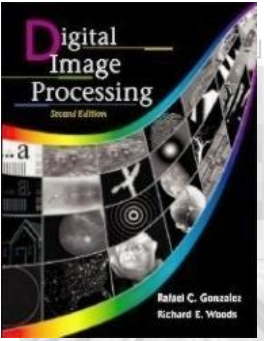
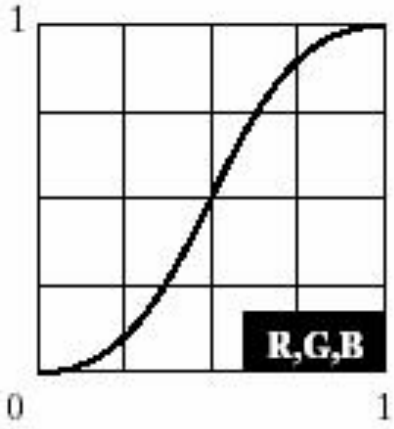


Tone and Color Corrections
Tone Corrections

FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not alter the image hues.

Flat

Corrected



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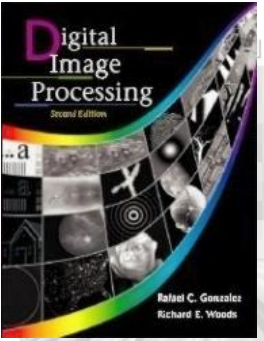
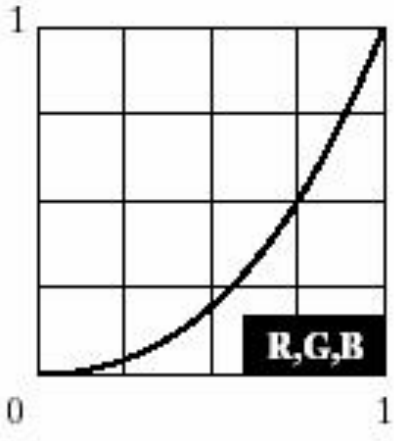


Tone and Color Corrections
Tone Corrections

FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not alter the image hues.

Light

Corrected



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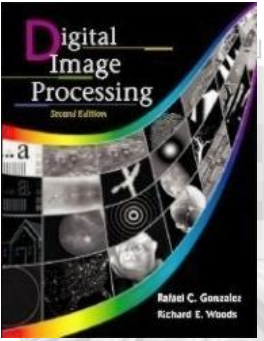
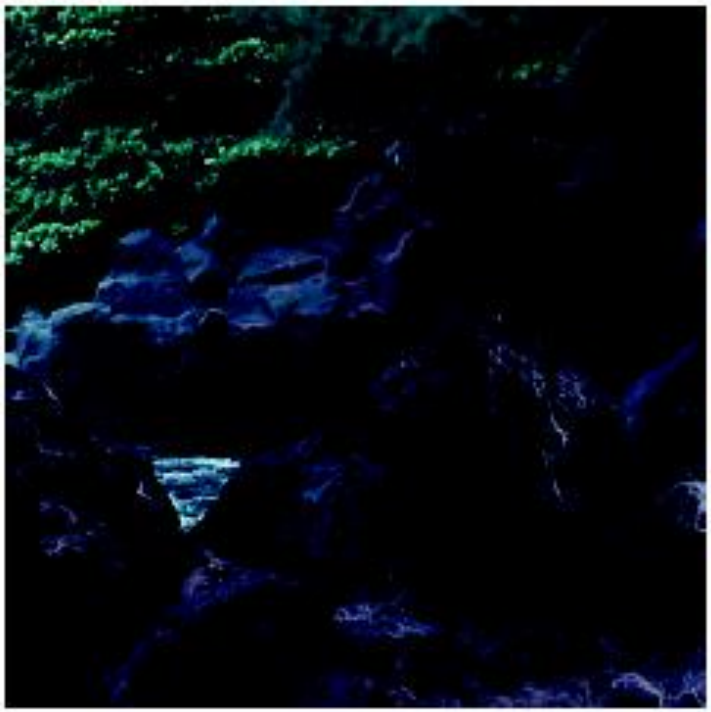


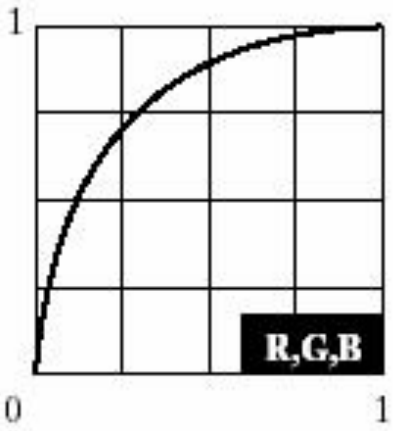
Tone and Color Corrections
Tone Corrections

FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not alter the image hues.



Dark

Corrected



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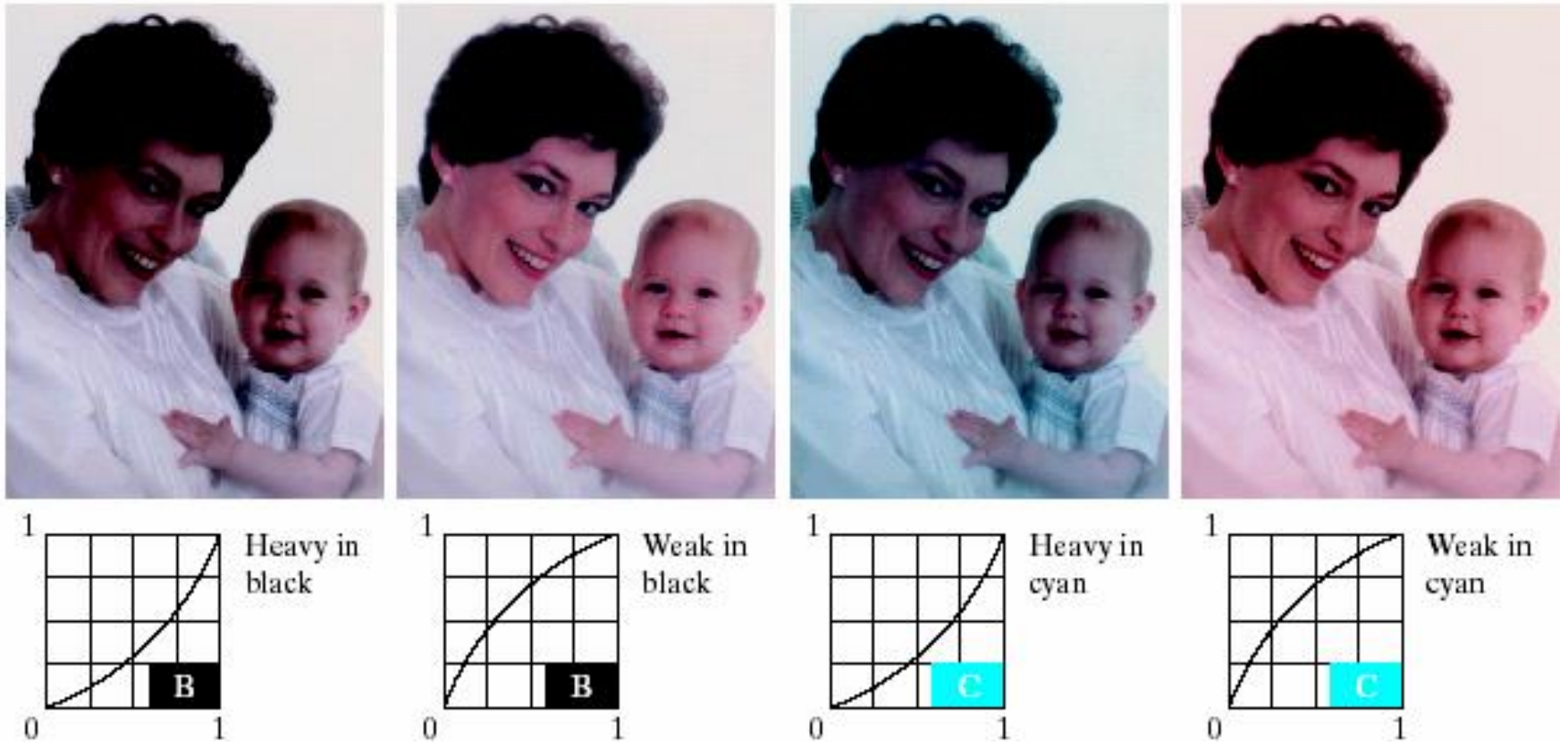
Tone and Color Corrections
Color Corrections

Original/Corrected

FIGURE 6.36 Color balancing corrections for CMYK color images.



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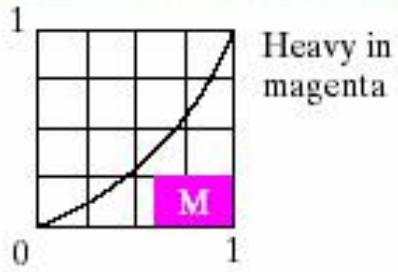
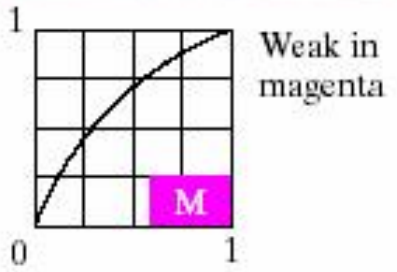
Tone and Color Corrections
Color Corrections

Original/Corrected

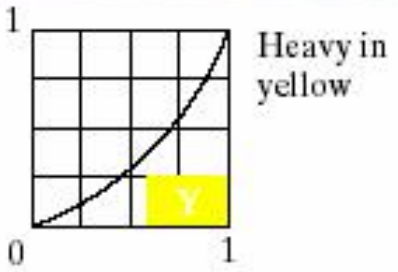
FIGURE 6.36 Color balancing corrections for CMYK color images.



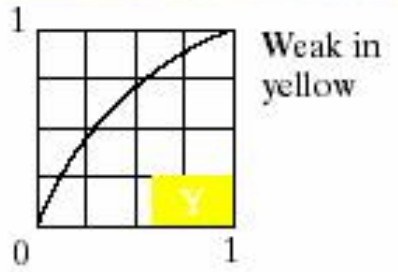
Heavy in magenta



Weak in magenta



Heavy in yellow



Weak in yellow

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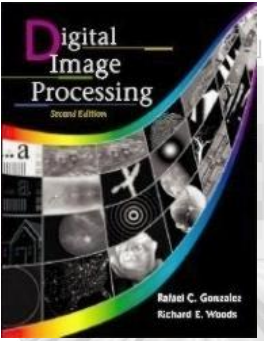

Smoothing and Sharpening
Color Image Smoothing

a b
c d

FIGURE 6.38
(a) RGB image. (b) Red component image. (c) Green component. (d) Blue component.

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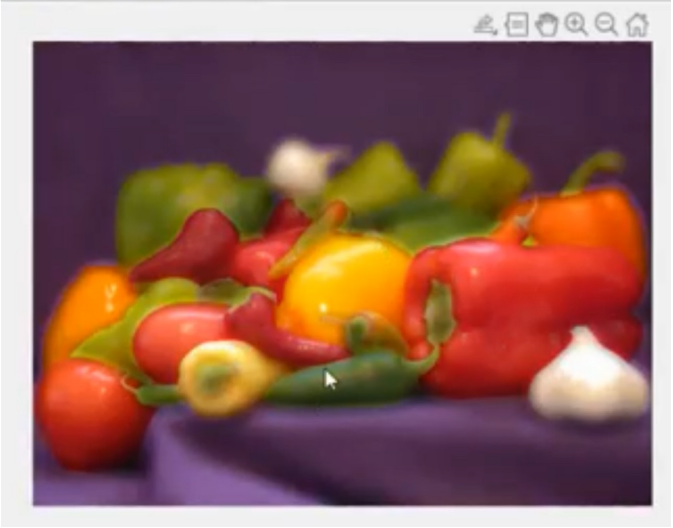
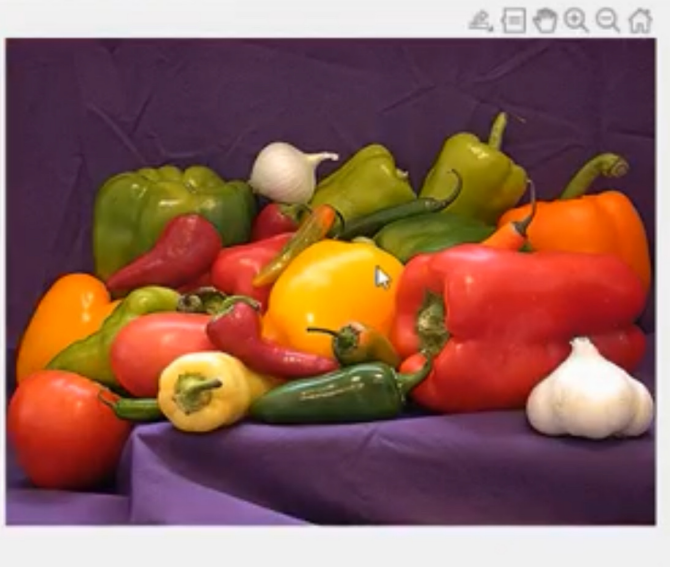
```
i=imread('peppers.png');
j=rgb2hsv(i);
imshow(i);figure;imshow(j);
h=j(:,:,1);
s=j(:,:,2);
v=j(:,:,3);
m=fspecial('average',15);
v_filtered=imfilter(v,m,'replicate');
j_pro=cat(3,h,s,v_filtered);
i_filtered=hsv2rgb(j_pro);
figure;imshow(i_filtered);
```



```
i=imread('peppers.png');
j=rgb2hsv(i);
imshow(i);figure;imshow(j);
h=j(:,:,1);
s=j(:,:,2);
v=j(:,:,3);
wl=fspecial('laplacian',0);
v_filtered=v-imfilter(v,wl,'replicate');
j_pro=cat(3,h,s,v_filtered);
i_filtered=hsv2rgb(j_pro);
figure;imshow(i_filtered);
```



Smoothing and Sharpening
Color Image Smoothing

a b c

FIGURE 6.39 HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.

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Smoothing and Sharpening
Color Image Smoothing

a b c

FIGURE 6.40 Image smoothing with a 5 × 5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

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Smoothing and Sharpening
Color Image Sharpening

- Laplacian method: see Section 3.7.2 (Pages 128-131)

a b c

FIGURE 6.41 Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the intensity component and converting to RGB. (c) Difference between the two results.

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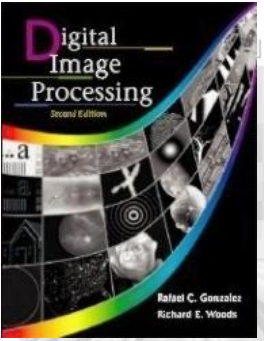
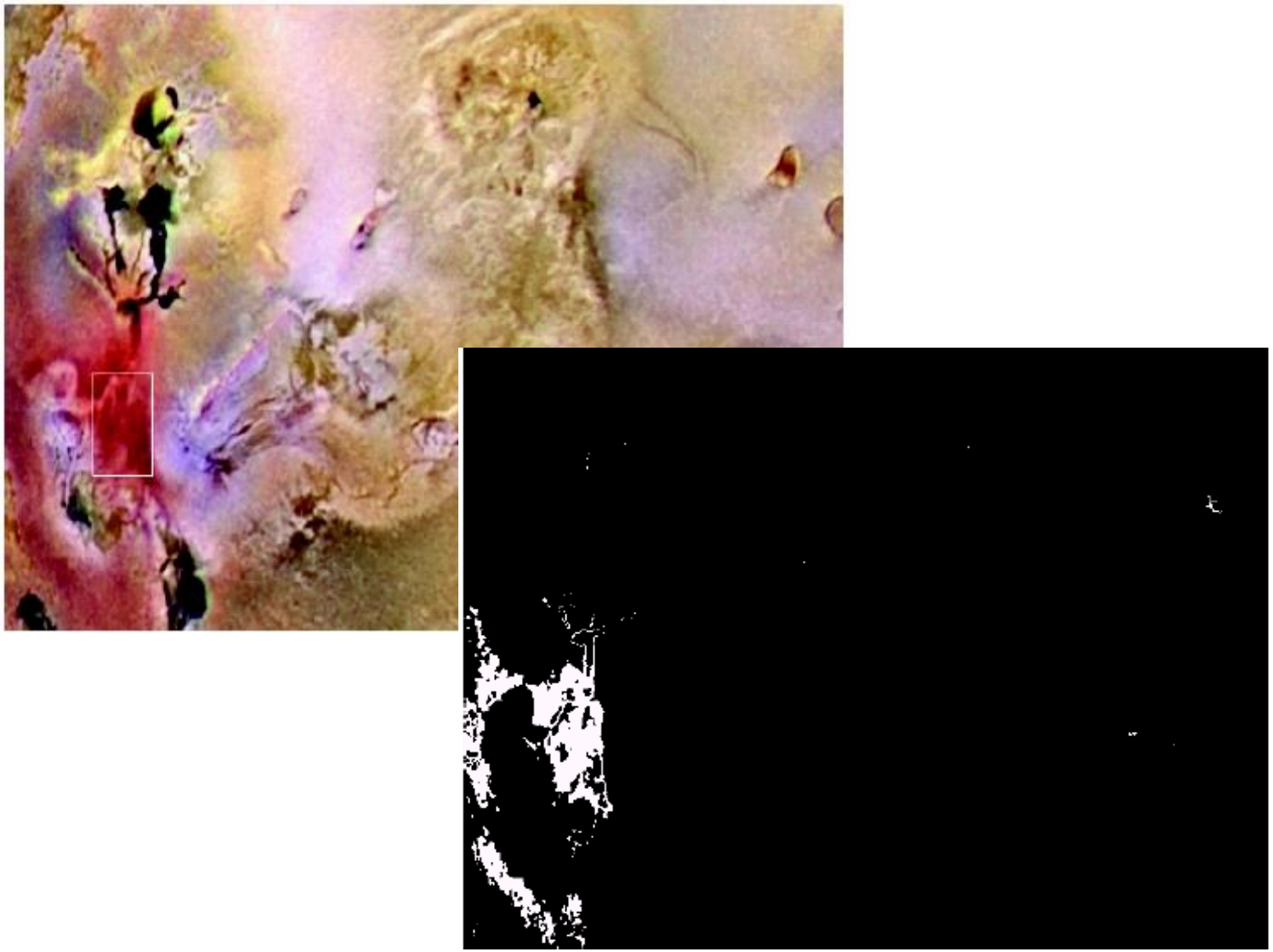


Image Segmentation
Segmentation in RGB Color Space



a
b

FIGURE 6.44
Segmentation in RGB space. (a) Original image with colors of interest shown enclosed by a rectangle. (b) Result of segmentation in RGB vector space. Compare with Fig. 6.42(h).

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